# Week 11 - Graded Assignment 11

**The due date for submitting this assignment has passed.**
Due on 2026-04-29, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

## You have last submitted on: 2026-04-29, 11:51 IST

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

*1 point*
Consider the following function

$$f(x, y, z) = x^2 z + y^3 e^{xy} + \sin(xyz).\text{f(x, y, z) = x2z + y3exy + sin(xyz).}$$

Which of the following option(s) is (are) true?

- $f_{xx} = 2z + y^5 e^{xy} - yz\sin(xyz)$fxx = 2z + y5exy − yz sin(xyz)
- $f_{xx} = 2z + y^5 e^{xy} - (yz)^2 \sin(xyz)$fxx = 2z + y5exy − (yz)2 sin(xyz)
- $f_{zy} = \cos(xyz)x^2 yz$fzy = cos(xyz)x2yz
- $f_{xz} = 2x - xy^2 z\sin(xyz)$fxz = 2x − xy2z sin(xyz)

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
$f_{xx} = 2z + y^5 e^{xy} - (yz)^2 \sin(xyz)$fxx = 2z + y5exy − (yz)2 sin(xyz)
Number of critical points of the function $f(x, y, z) = e^x (x^2 - y^2 - 2z^2)$
f(x, y, z) = ex(x2 − y2 − 2z2) is
2
**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
(Type: Numeric) 2
*1 point*
*1 point*
Choose the correct statement(s) about $f(x, y) = \sqrt{x^2 + y^2}$f(x, y) = x2 + y2
√
.

- $f$f has no critical points.
- $(0, 0)$(0, 0) is the only critical point of $f$f.
- $(0, 0)$(0, 0) is a local maximum of $f$f.
- $(0, 0)$(0, 0) is a local minimum of $f$f.

$(0, 0)$(0, 0) is a global minimum of $f$f.

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
$(0, 0)$(0, 0) is the only critical point of $f$f.
$(0, 0)$(0, 0) is a local minimum of $f$f.
$(0, 0)$(0, 0) is a global minimum of $f$f.

Find the minimum value of the surface area of a rectangular box that has a volume of 10001000 unit$^3$ 3.

Note:Note: If $x$, $y$x, y and $z$z represent the lengths of the three sides of a rectangular box, then the surface area and volume are given by $2(xy + yz + zx)$2(xy + yz + zx) and $xyz$xyz, respectively.
600

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
(Type: Numeric) 600
*1 point*
*1 point*

Consider a function $f(x, y, z) = x^3 + 3y^3 + z^3 - 4x - 6y - 4z$
f(x, y, z) = x3 + 3y3 + z3 − 4x − 6y − 4z. If $A$A is the the Hessian matrix of $f(x, y, z)$f(x, y, z) at (0,1,1), then which of the following option(s) is(are) true?

- [x] $det(A)$ = 0det(A) = 0
- [x] $Rank(A)$ = 2Rank(A) = 2
- [x] $Nullity(A)$ = 1Nullity(A) = 1
- [ ] $det(A)$ = 648det(A) = 648
- [ ] $Rank(A)$ = 3Rank(A) = 3

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
$det(A)$ = 0det(A) = 0
$Rank(A)$ = 2Rank(A) = 2
$Nullity(A)$ = 1Nullity(A) = 1
*1 point*

Consider the following function

$$f(x, y) = e^x + e^y - e^{x+y}$$ f(x, y) = ex + ey − ex+y

Which of the following option(s) is (are) true?

- [ ] Hessian matrix of $f$f at point $(1, 1)$(1, 1) is $\begin{bmatrix} 1-e & -e^2 \\ -e^2 & 1-e \end{bmatrix}$[1 − e−e2−e21 − e].
- [x] Hessian matrix of $f$f at point $(1, 1)$(1, 1) is $\begin{bmatrix} e-e^2 & -e^2 \\ -e^2 & e-e^2 \end{bmatrix}$[e − e2−e2−e2e − e2].
- [ ]

$f$f has a point of maximum.

$f$f has a saddle point.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

Hessian matrix of $f$f at point $(1, 1)(1, 1)$ is $\begin{bmatrix} e - e^2 & -e^2 \\ -e^2 & e - e^2 \end{bmatrix}$[e − e2−e2−e2e − e2].

$f$f has a saddle point.

*1 point*

Choose the correct statements.

If $f: R^3 \rightarrow R$f : R3 → R then the Hessian matrix $H_f(x, y, z)$Hf(x, y, z) is same for all $(x, y, z) \in R^3$(x, y, z) ∈ R3.

If $f: R^3 \rightarrow R$f : R3 → R is a linear transformation then the Hessian matrix $H_f(x, y, z)$Hf(x, y, z) is same for all $(x, y, z) \in R^3$(x, y, z) ∈ R3.

For each $n \in N$n ∈ N, there exists a map $f: R^n \rightarrow R$f : Rn → R such that the rank of the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$Hf(x1, x2, … , xn) is $n$n for some $(x_1, x_2, \ldots, x_n) \in R^n$(x1, x2, … , xn ) ∈ Rn.

For each $n \in N$n ∈ N, there exists a map $f: R^n \rightarrow R$f : Rn → R such that the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$Hf(x1, x2, … , xn) is symmetric for all $(x_1, x_2, \ldots, x_n) \in R^n$(x1, x2, … , xn ) ∈ Rn.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

If $f: R^3 \rightarrow R$f : R3 → R is a linear transformation then the Hessian matrix $H_f(x, y, z)$Hf(x, y, z) is same for all $(x, y, z) \in R^3$(x, y, z) ∈ R3.

For each $n \in N$n ∈ N, there exists a map $f: R^n \rightarrow R$f : Rn → R such that the rank of the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$Hf(x1, x2, … , xn) is $n$n for some $(x_1, x_2, \ldots, x_n) \in R^n$(x1, x2, … , xn ) ∈ Rn.

For each $n \in N$n ∈ N, there exists a map $f: R^n \rightarrow R$f : Rn → R such that the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$Hf(x1, x2, … , xn) is symmetric for all $(x_1, x_2, \ldots, x_n) \in R^n$(x1, x2, … , xn ) ∈ Rn.

The maximum value of the function $-2x^2 + 2xy - y^2 + 12x - 4y - 7$−2x2 + 2xy − y2 + 12x − 4y − 7 is

13

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) 13

*1 point*

*1 point*

Define a function

$$f(x, y)\text{f(x, y)} = \begin{cases} (x - y)^2 \sin\left(\frac{1}{x - y}\right) & \text{if } x \neq y \\ 0 & \text{if } x = y. \end{cases}\{\text{(x − y)2 sin(x−y1)0if x} \quad = \text{yif x = y.}$$

Choose the set of correct options.

$f$ f is continuous at $(0,0)$(0, 0).

$f_x(0,0) \neq f_y(0,0)$fx(0, 0) = fy(0, 0)

$f_x(0,0) = f_y(0,0)$fx(0, 0) = fy(0, 0)

$f$ f is differentiable at $(0,0)$(0, 0).

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
$f$ f is continuous at $(0,0)$(0, 0).
$f_x(0,0) = f_y(0,0)$fx(0, 0) = fy(0, 0)
$f$ f is differentiable at $(0,0)$(0, 0).

A company produces two types of chocolates, one that sells for Rs.3 and another at Rs.22. The total revenue, in lakhs, from the sale of $x$x lakhs chocolates at Rs.22 each and $y$y lakhs at Rs.33 each is given by

$$F(x, y) = 2x + 3y.\text{F}(\text{x}, \text{y}) = 2\text{x} + 3\text{y}.$$

The total cost in lakhs for producing $x$x lakhs of the Rs.22 chocolates and $y$y lakhs of the Rs.33 chocolates is given by

$$G(x, y) = 2x^2 - 2xy + y^2 - 9x + 6y + 7.\text{G}(\text{x}, \text{y}) = 2\text{x}2 - 2\text{xy} + \text{y}2 - 9\text{x} + 6\text{y} + 7.$$

Then the maximum profit in lakhs is

[Enter your answer upto two decimal places]
11.25
**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
(Type: Numeric) 11.25
*1 point*

*1 point*
(MSQ) Let $f: R^2 \rightarrow R$f : R2 → R be a multivariable function and $(x_0, y_0) \in R^2$(x0, y0) ∈ R2. Consider the following statements:

- P: $f$ f is continuous at $(x_0, y_0)$(x0, y0).
- Q: The partial derivatives of $f$ f exist and are continuous in a neighborhood of $(x_0, y_0)$(x0, y0).
- R: The tangent plane to the graph of $f$ f exists at $(x_0, y_0)$(x0, y0).

Choose the correct statements from the following.

P implies that $f$ f is differentiable at $(x_0, y_0)$(x0, y0).

If $f$ f is differentiable at $(x_0, y_0)$(x0, y0), then P is satisfied.

Q implies that $f$ f is differentiable at $(x_0, y_0)$(x0, y0).

☐ If $f$ f is differentiable at $(x_0, y_0)$(x0, y0), then Q is satisfied.

☑ R implies that $f$ f is differentiable at $(x_0, y_0)$(x0, y0).

☑ If $f$ f is differentiable at $(x_0, y_0)$(x0, y0), then R is satisfied.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

If $f$ f is differentiable at $(x_0, y_0)$(x0, y0), then P is satisfied.
Q implies that $f$ f is differentiable at $(x_0, y_0)$(x0, y0).
R implies that $f$ f is differentiable at $(x_0, y_0)$(x0, y0).
If $f$ f is differentiable at $(x_0, y_0)$(x0, y0), then R is satisfied.

(NAT) A particle moving in a potential field tends to settle at points where the potential energy is a local minimum (stable equilibrium point). Consider a particle moving in $R^2$ R2 where the potential energy as a function of position is given by

$$V(x, y) = x^4 + y^4 - 4xy + 1 \text{V (x, y) = x4 + y4 − 4xy + 1,}$$

for all $(x, y) \in R^2$ (x, y) ∈ R2. Using the Hessian test, find the potential energy at stable equilibrium points.

-1

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) -1

*1 point*